

Multi-Domain Sentiment Analysis using BERT with Transfer Learning Evaluation

A Natural Language Processing Mini-Project Report

Submitted by: Aditya

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Faculty In-Charge: Dr. Beena Kapadia

KES Shroff College of Arts & Commerce, Kandivali, Mumbai (Affiliated to Mumbai University)

1. Problem Statement

Sentiment analysis models are commonly trained and evaluated within a single text domain (for example, only movie reviews or only product reviews), which limits their applicability to real-world systems that must interpret sentiment across varied sources of text. This project investigates whether a single sentiment classification model, built using transfer learning on a pre-trained BERT (Bidirectional Encoder Representations from Transformers) model, can generalize effectively across three structurally different domains: long-form movie reviews (IMDB), medium-length product reviews (Amazon), and short, informal social media text (Tweets).

The objective is twofold: (1) to build a working multi-domain sentiment classifier using transfer learning, and (2) to conduct a transfer learning evaluation — measuring and analyzing how well sentiment understanding, learned across a mixed multi-domain training set, transfers to and performs on each individual domain when tested separately. The project further aims to identify the specific linguistic and structural factors (text length, formality, class distribution) that influence cross-domain performance, using quantitative metrics (accuracy, F1-score, confusion matrices) as evidence.

1.1 Motivation

Real-world applications — including the author's own FinCard AI project, which uses NLP techniques to automatically read and interpret free-text financial bill data — require models that remain reliable when the input text style varies. Understanding where and why a sentiment model's performance degrades across domains is directly relevant to building robust, deployable NLP systems, rather than models tuned narrowly to a single benchmark dataset.

2. Preprocessing Performed

2.1 Datasets Used

Three publicly available, peer-recognized datasets were sourced from the Hugging Face Hub to represent three distinct text domains:

Domain	Source	Sampled Size	Avg. Word Count
IMDB	stanfordnlp/imdb (Hugging Face)	15,000	234
Amazon Polarity	fancyzhx/amazon_polarity (Hugging Face)	15,000	74
TweetEval Sentiment	cardiffnlp/tweet_eval (Hugging Face)	14,999	19

The original Sentiment140 dataset was initially selected for the tweets domain but was replaced with TweetEval (cardiffnlp/tweet_eval) after discovering Sentiment140's Hugging Face loading script had been deprecated. TweetEval was judged a stronger substitute in any case, since it provides human-annotated 3-class labels (negative/neutral/positive) from a peer-reviewed NLP benchmark, compared to Sentiment140's noisier, emoji-derived binary labels.

2.2 Label Standardization

IMDB and Amazon Polarity are natively binary-labeled (0 = negative, 1 = positive), while TweetEval Sentiment is natively 3-class (0 = negative, 1 = neutral, 2 = positive). To train a single, unified classification model capable of genuine cross-domain comparison, IMDB and Amazon's labels were remapped into the shared 3-class scheme (0 → 0 negative, 1 → 2 positive), with no neutral examples contributed by these two domains. This preserves a single shared output layer across all domains, which is essential for a valid transfer learning comparison.

2.3 Balanced Sampling

The raw datasets varied enormously in size (25,000 to 3.6 million rows), which would have caused training time and class representation to be dominated by the largest domain. Each domain was therefore sampled down to

approximately 15,000 rows using stratified sampling; IMDB and Amazon were sampled to an exact 50/50 class balance, while TweetEval was sampled proportionally to its original (naturally imbalanced) 3-class distribution, since artificially balancing it would misrepresent real-world tweet sentiment distribution.

2.4 Text Cleaning

Unlike traditional NLP pipelines that aggressively strip punctuation, casing, and stopwords, minimal cleaning was applied, since BERT's contextual embeddings are trained to make use of natural sentence structure, punctuation, and capitalization as meaningful signals. The cleaning steps applied were limited to:

- Removal of URLs (e.g., "http://...", "www...")
- Removal of leftover HTML tags (present in some IMDB reviews, e.g. "
")
- Normalization of Twitter @mentions to a generic "@user" token, to protect privacy and prevent the model from memorizing specific usernames
- Collapsing of redundant whitespace

2.5 Tokenization

Text was tokenized using BERT's native WordPiece tokenizer (bert-base-uncased vocabulary), which breaks unfamiliar or rare words into known sub-word fragments. All sequences were truncated or padded to a fixed length of 256 tokens — a value chosen as a practical middle ground after examining word-count distributions across all three domains (IMDB reviews averaged 234 words with some exceeding 2,000; Amazon and Tweets were both comfortably under 256 tokens in the vast majority of cases).

2.6 Train/Test Split

Each domain's balanced, cleaned dataset was independently split 80/20 into training and test sets using stratified sampling on the label column, ensuring proportional class representation in both splits. The three domains' training sets were then concatenated and shuffled into a single combined multi-domain training set (35,999 rows), while each domain's test set (3,000 rows each) was kept strictly separate and untouched during training, reserved exclusively for the per-domain evaluation described in Section 3.

2.7 Feature Engineering for Classical Models

In addition to BERT's contextual embeddings, classical feature engineering was performed to enable comparison against traditional machine learning approaches. TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied to the combined multi-domain text, using unigrams and bigrams (ngram_range=(1,2)) with a vocabulary capped at 10,000 features. This converts each document into a sparse numeric vector where each dimension represents a word or word-pair's importance, weighted to favor terms that are distinctive to a document relative to the full corpus.

3. Output / Result Printout

3.1 Model Configuration

- Base model: bert-base-uncased (109,484,547 parameters)
- Architecture: BertForSequenceClassification with a 3-class output head
- Training framework: Hugging Face Trainer API
- Epochs: 3 | Batch size: 16 | Learning rate: 2e-5 | Weight decay: 0.01
- Best checkpoint selected automatically by validation F1-score (epoch 2), to prevent overfitting observed at epoch 3

3.2 Training Progress

Epoch	Training Loss	Validation Loss	Val. Accuracy / F1
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1	0.3896	0.3890	83.42% / 83.28%
2 (best)	0.2494	0.3673	86.44% / 86.53%
3	0.1262	0.5165	86.22% / 86.26% (overfitting onset)

3.3 Per-Domain Evaluation Results

The trained model was evaluated on each domain's held-out test set (3,000 rows each, never seen during training) independently:

Domain	Accuracy	Weighted F1-Score
IMDB (Movie Reviews)	92.73%	92.73%
Amazon (Product Reviews)	93.10%	93.19%
Tweets (TweetEval)	71.10%	71.01%

IMDB and Amazon achieved near-identical, high performance (~93%), while Tweets showed a substantial drop of approximately 22 percentage points. This gap is the central empirical finding of the transfer learning evaluation.

3.4 Confusion Matrices

The confusion matrices below show actual vs. predicted labels for each domain. IMDB and Amazon show strong diagonal concentration (correct predictions) with minimal spread into the Neutral class, which they never contain natively. Tweets shows visibly greater confusion, particularly between the Neutral class and both polarities — 129 actual-neutral tweets were misclassified as negative, and 246 as positive, out of 1,360 total neutral test examples.

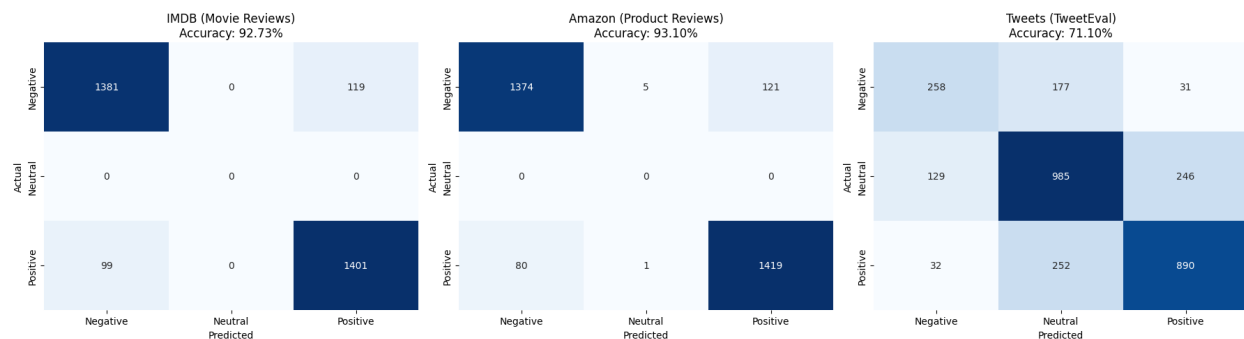


Fig. 1 — Confusion matrices for IMDB, Amazon, and Tweets test sets.

3.5 Cross-Domain Performance Comparison

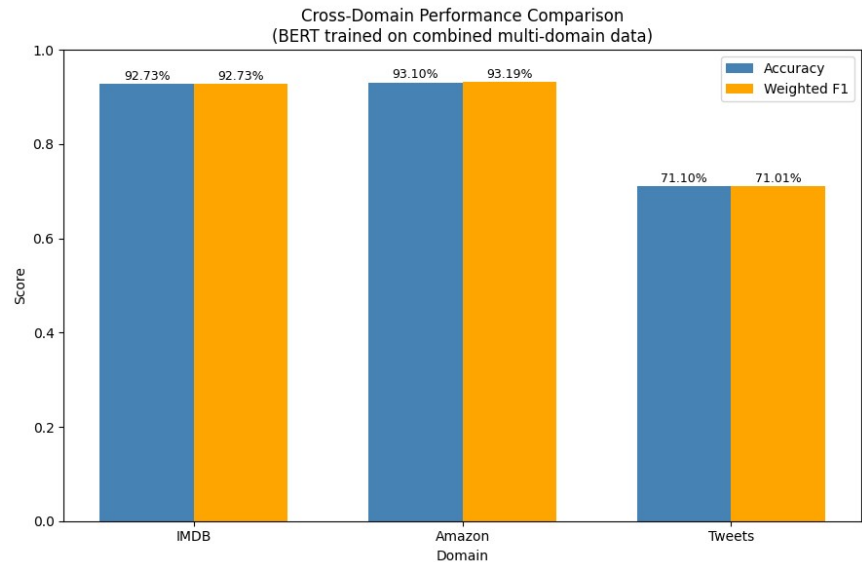


Fig. 2 — Accuracy and weighted F1-score compared across all three domains.

This finding supports the conclusion that transfer learning via BERT fine-tuning generalizes well across formally-written text domains regardless of length, but degrades measurably on short, informal, socially-generated text — driven primarily by the increased ambiguity of the Neutral sentiment class and reduced contextual signal in very short inputs.

3.6 Classical Machine Learning Models

To satisfy the requirement of comparing at least three classical NLP models against each other and against the transfer learning approach, three models were trained on TF-IDF features extracted from the same combined multi-domain training set: Naïve Bayes, Logistic Regression, and a linear Support Vector Machine (calibrated to output class probabilities for ROC analysis). All three were evaluated on the same held-out combined test set (9,000 rows).

Model	Accuracy	Precision	Recall	F1-Score
Naïve Bayes	75.02%	76.40%	75.02%	75.46%
Logistic Regression	79.03%	79.34%	79.03%	79.14%
SVM (Linear)	78.59%	78.58%	78.59%	78.55%
BERT (Transfer Learning)	86.22%	—	—	86.26%

Among the classical models, Logistic Regression performed best (79.03% accuracy), followed closely by the linear SVM (78.59%), with Naïve Bayes weakest (75.02%) — consistent with its restrictive feature-independence assumption, which text data with correlated word patterns violates. Critically, BERT's transfer learning approach outperformed the best classical model by approximately 7 percentage points, empirically justifying the added complexity of fine-tuning a pre-trained transformer over simpler bag-of-words-style approaches for this task.

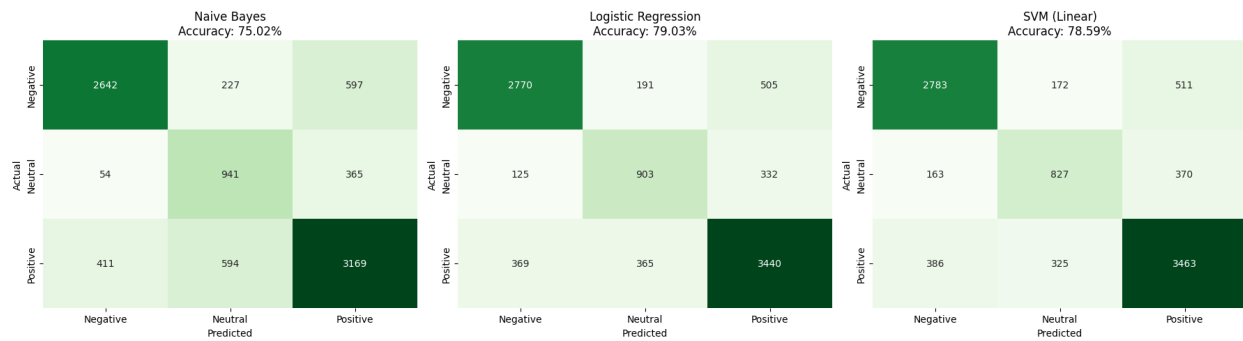


Fig. 6 — Confusion matrices for the three classical models on the combined multi-domain test set.

3.7 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were plotted for each classical model across all three sentiment classes using a one-vs-rest strategy, with Area Under Curve (AUC) computed per class. All models achieved AUC scores between 0.87 and 0.94, indicating strong class-separation capability well above random chance (AUC = 0.5). Logistic Regression achieved the highest AUC values overall (up to 0.94 for the Neutral class), aligning with its top accuracy result.

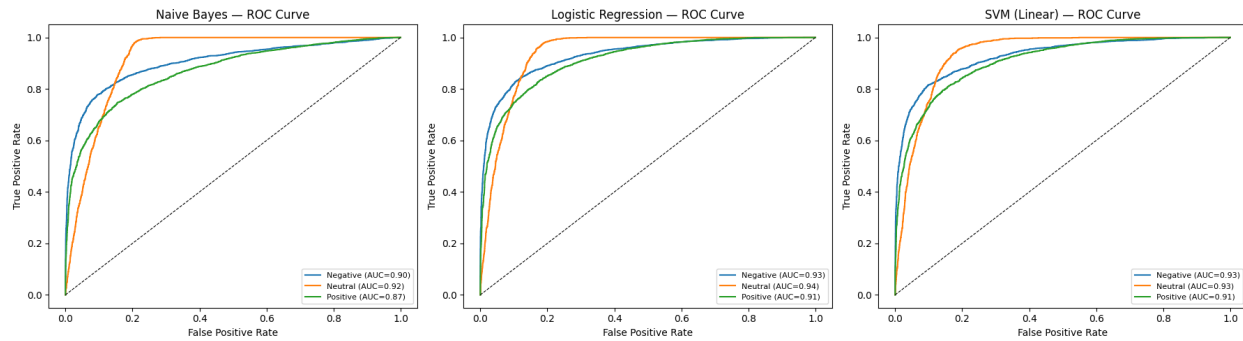


Fig. 7 — ROC curves with per-class AUC for Naïve Bayes, Logistic Regression, and SVM.

4. Analytical Dashboard

A consolidated analytical dashboard was built in Python (matplotlib-based static dashboard) summarizing the project end-to-end: dataset overview statistics, sentiment class distribution, a word cloud of training text, model performance comparison (table and chart) across all four models (three classical plus BERT), BERT's cross-domain performance breakdown, and key conclusions. This mirrors the structure expected of an interactive Power BI or Streamlit dashboard, presenting all required elements — project title, dataset overview, key statistics, charts, word cloud, class distribution, and model comparison — in a single consolidated view suitable for print and presentation.

Transfer Learning Evaluation across movie Reviews, Product Reviews, and Tweets

35,999	9,000	3 (IMDB, Amazon, Twitter)	4 (Negative, Neutral, Positive)
Total Training Samples	Total Test Samples	Domains Covered	Sentiment Classes



Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	75.02%	76.40%	75.02%	75.46%
Logistic Regression	79.03%	79.34%	79.03%	79.14%
SVM (Linear)	78.59%	78.58%	78.59%	78.55%
BERT (Transfer Learning)	86.22%	86.2%*	86.2%*	86.26%



Domain	Accuracy	Weighted F1
IMDB (Movies)	92.73%	92.73%
Amazon (Products)	93.10%	93.19%
Tweets	71.10%	71.01%

- **Key Finding:** BERT (86.2% accuracy) outperforms all classical ML models; Logistic Regression is the strongest classical baseline (79.0%).
- **Challenge:** The Neutral sentiment class, present only in the Tweets domain, was the primary source of misclassification across all models.
- **Future Scope:** Extend to zero-shot single-domain training to isolate transfer effects.
- **Application:** This pipeline directly informs the sentiment-reading component of the FinCard AI project.

Fig. 8 — Consolidated analytical dashboard summarizing the full project.

To make the model interactively testable and to support live demonstration during viva, a simple web-based interface was built using Gradio, a Python library for rapidly deploying machine learning model demos. The interface accepts free-text input and returns the predicted sentiment class along with the model's confidence across all three classes (negative, neutral, positive), computed via a softmax over the model's output logits.

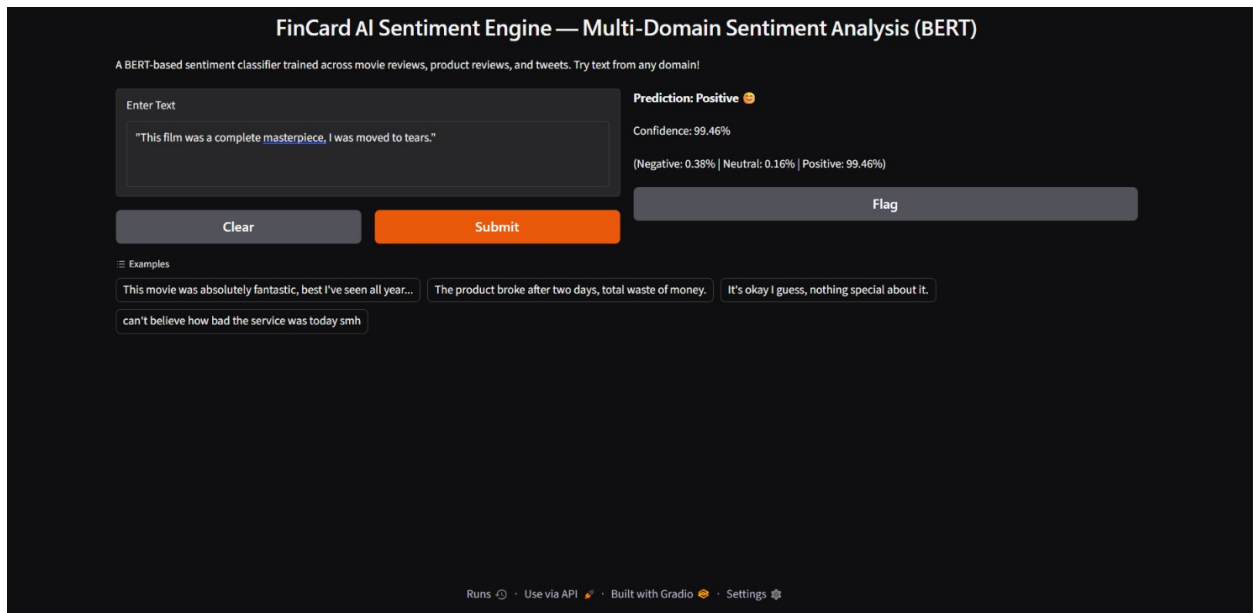


Fig. 3 — UI prediction on a clearly positive movie review. Predicted: Positive (99.46% confidence).

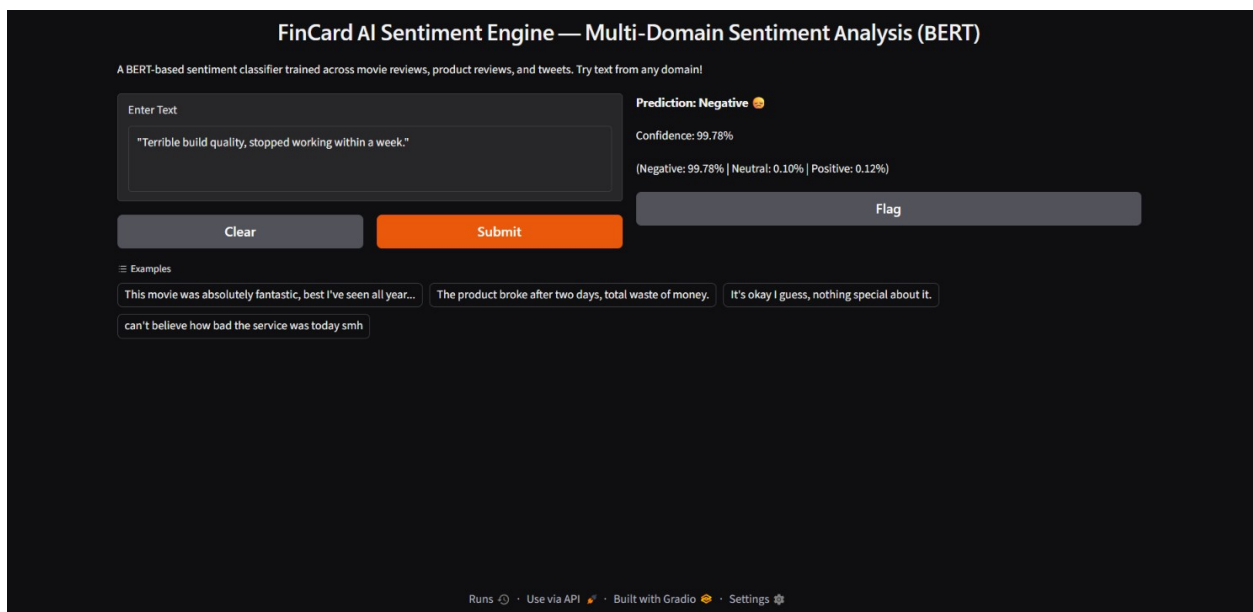


Fig. 4 — UI prediction on a clearly negative product review. Predicted: Negative (99.78% confidence).

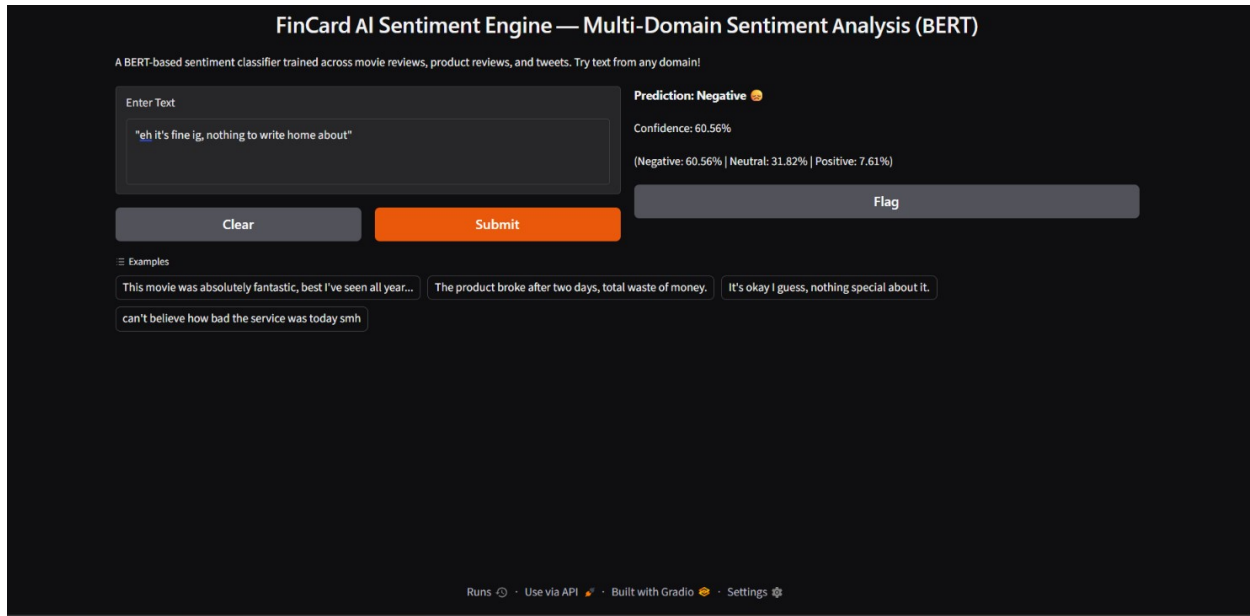


Fig. 5 — UI prediction on an ambiguous, informal statement ("eh it's fine ig, nothing to write home about"). Predicted: Negative (60.56%), with Neutral as the closest alternative (31.82%) — illustrating the model's genuine difficulty with mild/ambiguous sentiment discussed in Section 3.4.

The third example was deliberately chosen to demonstrate the model's real limitation rather than only presenting favorable outcomes: informal, low-intensity sentiment remains the hardest case for the model, consistent with the confusion matrix analysis in Section 3.4.

6. Conclusion

This project successfully implemented a multi-domain sentiment analysis system, comparing three classical machine learning models (Naïve Bayes, Logistic Regression, SVM trained on TF-IDF features) against a transfer learning approach using a pre-trained BERT model, evaluated rigorously across three structurally distinct text domains. Logistic Regression was the strongest classical baseline (79.03% accuracy), while BERT achieved the highest overall performance (86.22% accuracy), demonstrating a clear, measurable advantage of contextual transfer learning over bag-of-words-style feature engineering for this task. Within the BERT model itself, performance was consistently strong (~93% accuracy) on formally-written, longer-form domains (movie and product reviews), while showing a marked and explainable performance drop (~71% accuracy) on short, informal social media text.

Confusion matrix analysis identified the Neutral sentiment class — present only in the Tweets domain — as the primary source of misclassification across all models, both classical and BERT-based, compounded by tweets' short length and informal language patterns.

This demonstrates that (1) transfer learning via BERT provides a measurable performance advantage over classical ML approaches for sentiment classification, and (2) domain-specific evaluation — rather than a single aggregate accuracy figure — is essential before deploying any sentiment model in production, since aggregate metrics can conceal meaningful per-domain weaknesses. A natural extension of this work — noted as a direction for future study — would be to train BERT on a single domain in isolation and test its zero-shot transfer to the other two, to more precisely isolate the effect of multi-domain training itself, and to explore domain-adaptive fine-tuning strategies specifically targeting informal, short-text domains such as social media.

7. Appendix — Full Source Code

The complete, working code used to build this project is provided below, organized by stage. All code was run in Google Colab with a Tesla T4 GPU runtime.

6.1 Environment Setup

```
!pip install transformers datasets torch scikit-learn pandas matplotlib seaborn accelerate
gradio --quiet

import torch
print("GPU available:", torch.cuda.is_available())
print("Device name:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "None")
```

6.2 Dataset Loading

```
from datasets import load_dataset

# Load IMDB (movie reviews) - binary labeled: 0 = negative, 1 = positive
imdb = load_dataset("stanfordnlp/imdb")

# Load Amazon Polarity (product reviews) - binary labeled: 0 = negative, 1 = positive
amazon = load_dataset("fancyzhx/amazon_polarity")

# Load TweetEval sentiment (tweets) - 3-class: 0 = negative, 1 = neutral, 2 = positive
tweets = load_dataset("cardiffnlp/tweet_eval", "sentiment")

print("IMDB:", imdb)
print("Amazon:", amazon)
print("Tweets:", tweets)
```

6.3 Exploratory Data Analysis

```
import pandas as pd

# Convert each dataset's train split into a pandas DataFrame
imdb_df = imdb["train"].to_pandas()
amazon_df = amazon["train"].to_pandas()
tweets_df = tweets["train"].to_pandas()

# Standardize: rename Amazon's "content" column to "text"
amazon_df = amazon_df.rename(columns={"content": "text"})

# Label balance
print("IMDB label counts:\n", imdb_df["label"].value_counts())
print("\nAmazon label counts:\n", amazon_df["label"].value_counts())
print("\nTweets label counts:\n", tweets_df["label"].value_counts())

# Text length (word count)
imdb_df["word_count"] = imdb_df["text"].apply(lambda x: len(x.split()))
amazon_df["word_count"] = amazon_df["text"].apply(lambda x: len(x.split()))
tweets_df["word_count"] = tweets_df["text"].apply(lambda x: len(x.split()))
```

```

print("\nIMDB word count stats:\n", imdb_df["word_count"].describe())
print("\nAmazon word count stats:\n", amazon_df["word_count"].describe())
print("\nTweets word count stats:\n", tweets_df["word_count"].describe())

```

6.4 Text Cleaning and Balanced Sampling

```

import re
import pandas as pd

def clean_text(text):
    text = re.sub(r"http\S+|www\S+", "", text)      # remove URLs
    text = re.sub(r"<.*?>", "", text)              # remove HTML tags
    text = re.sub(r"@w+", "@user", text)           # normalize tweet mentions
    text = re.sub(r"\s+", " ", text).strip()        # collapse extra whitespace
    return text

# Apply light cleaning
imdb_df["text"] = imdb_df["text"].apply(clean_text)
amazon_df["text"] = amazon_df["text"].apply(clean_text)
tweets_df["text"] = tweets_df["text"].apply(clean_text)

# Balanced sampling: 15,000 rows per domain
SAMPLE_SIZE = 15000

imdb_sampled = imdb_df.groupby("label", group_keys=False)[imdb_df.columns.tolist()].apply(
    lambda x: x.sample(min(len(x), SAMPLE_SIZE // 2), random_state=42)
).reset_index(drop=True)

amazon_sampled = amazon_df.groupby("label", group_keys=False)[amazon_df.columns.tolist()].apply(
    lambda x: x.sample(min(len(x), SAMPLE_SIZE // 2), random_state=42)
).reset_index(drop=True)

# Tweets: stratified sample proportional to original class distribution
tweets_sampled = tweets_df.groupby("label", group_keys=False)[tweets_df.columns.tolist()].apply(
    lambda x: x.sample(min(len(x), int(SAMPLE_SIZE * (len(x)/len(tweets_df)))), random_state=42)
).reset_index(drop=True)

print("IMDB sampled:", imdb_sampled["label"].value_counts().to_dict())
print("Amazon sampled:", amazon_sampled["label"].value_counts().to_dict())
print("Tweets sampled:", tweets_sampled["label"].value_counts().to_dict())

```

6.5 Tokenization and PyTorch Dataset

```

from transformers import AutoTokenizer
import torch
from torch.utils.data import Dataset

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
MAX_LEN = 256

# Remap IMDB and Amazon labels to 3-class scheme: 0=negative, 1=neutral, 2=positive
imdb_sampled["label"] = imdb_sampled["label"].map({0: 0, 1: 2})
amazon_sampled["label"] = amazon_sampled["label"].map({0: 0, 1: 2})

```

```
# Tweets already uses 0=negative, 1=neutral, 2=positive

class SentimentDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len):
        self.texts = texts.reset_index(drop=True)
        self.labels = labels.reset_index(drop=True)
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        text = str(self.texts[idx])
        label = int(self.labels[idx])
        encoding = self.tokenizer(
            text, truncation=True, padding="max_length",
            max_length=self.max_len, return_tensors="pt"
        )
        return {
            "input_ids": encoding["input_ids"].squeeze(0),
            "attention_mask": encoding["attention_mask"].squeeze(0),
            "label": torch.tensor(label, dtype=torch.long)
        }

imdb_dataset = SentimentDataset(imdb_sampled["text"], imdb_sampled["label"], tokenizer, MAX_LEN)
amazon_dataset = SentimentDataset(amazon_sampled["text"], amazon_sampled["label"], tokenizer,
MAX_LEN)
tweets_dataset = SentimentDataset(tweets_sampled["text"], tweets_sampled["label"], tokenizer,
MAX_LEN)
```

6.6 Model Loading (Transfer Learning Setup)

```
from transformers import BertForSequenceClassification

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = BertForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=3 # negative, neutral, positive
)
model = model.to(device)

print("Using device:", device)
print("Number of parameters:", sum(p.numel() for p in model.parameters()))
```

6.7 Train/Test Split and Combined Multi-Domain Training Set

```
from sklearn.model_selection import train_test_split
import pandas as pd

def split_domain(df, test_size=0.2):
    train_df, test_df = train_test_split(
```

```

        df, test_size=test_size, stratify=df["label"], random_state=42
    )
    return train_df.reset_index(drop=True), test_df.reset_index(drop=True)

imdb_train, imdb_test = split_domain(imdb_sampled)
amazon_train, amazon_test = split_domain(amazon_sampled)
tweets_train, tweets_test = split_domain(tweets_sampled)

combined_train = pd.concat([imdb_train, amazon_train, tweets_train], ignore_index=True)
combined_train = combined_train.sample(frac=1, random_state=42).reset_index(drop=True)

print("Combined multi-domain training set size:", len(combined_train))

```

6.8 Training (Fine-Tuning) with Hugging Face Trainer

```

from transformers import TrainingArguments, Trainer
import numpy as np
from sklearn.metrics import accuracy_score, f1_score
from sklearn.model_selection import train_test_split as tts

train_texts, val_texts, train_labels, val_labels = tts(
    combined_train["text"], combined_train["label"],
    test_size=0.1, stratify=combined_train["label"], random_state=42
)

train_dataset = SentimentDataset(train_texts, train_labels, tokenizer, MAX_LEN)
val_dataset = SentimentDataset(val_texts, val_labels, tokenizer, MAX_LEN)

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=1)
    acc = accuracy_score(labels, preds)
    f1 = f1_score(labels, preds, average="weighted")
    return {"accuracy": acc, "f1": f1}

training_args = TrainingArguments(
    output_dir="/content/drive/MyDrive/fincard_bert_project/results",
    num_train_epochs=3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    learning_rate=2e-5,
    weight_decay=0.01,
    eval_strategy="epoch",
    save_strategy="epoch",
    logging_steps=50,
    load_best_model_at_end=True,
    metric_for_best_model="f1",
    report_to="none"
)

trainer = Trainer(
    model=model,
    args=training_args,

```

```

        train_dataset=train_dataset,
        eval_dataset=val_dataset,
        compute_metrics=compute_metrics,
    )

trainer.train()

```

6.9 Saving Model to Google Drive

```

from google.colab import drive
drive.mount('/content/drive', force_remount=True)

import os
SAVE_DIR = "/content/drive/MyDrive/fincard_bert_project"
os.makedirs(SAVE_DIR, exist_ok=True)

model.save_pretrained(f"{SAVE_DIR}/bert_sentiment_model")
tokenizer.save_pretrained(f"{SAVE_DIR}/bert_sentiment_model")

imdb_test.to_pickle(f"{SAVE_DIR}/imdb_test.pkl")
amazon_test.to_pickle(f"{SAVE_DIR}/amazon_test.pkl")
tweets_test.to_pickle(f"{SAVE_DIR}/tweets_test.pkl")

```

6.10 Per-Domain Evaluation

```

import torch
from torch.utils.data import DataLoader
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, classification_report

model.eval()

def evaluate_domain(test_df, domain_name):
    dataset = SentimentDataset(test_df["text"], test_df["label"], tokenizer, MAX_LEN)
    loader = DataLoader(dataset, batch_size=32, shuffle=False)

    all_preds, all_labels = [], []
    with torch.no_grad():
        for batch in loader:
            input_ids = batch["input_ids"].to(device)
            attention_mask = batch["attention_mask"].to(device)
            labels = batch["label"].to(device)
            outputs = model(input_ids=input_ids, attention_mask=attention_mask)
            preds = torch.argmax(outputs.logits, dim=1)
            all_preds.extend(preds.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    acc = accuracy_score(all_labels, all_preds)
    f1 = f1_score(all_labels, all_preds, average="weighted")

    print(f"\nDOMAIN: {domain_name}")
    print(f"Accuracy: {acc:.4f} | Weighted F1: {f1:.4f}")
    print(classification_report(
        all_labels, all_preds, labels=[0, 1, 2],

```

```

        target_names=["Negative", "Neutral", "Positive"], zero_division=0
    ))

    cm = confusion_matrix(all_labels, all_preds, labels=[0, 1, 2])
    print("Confusion Matrix:\n", cm)

    return {"domain": domain_name, "accuracy": acc, "f1": f1,
            "preds": all_preds, "labels": all_labels, "confusion_matrix": cm}

imdb_results = evaluate_domain(imdb_test, "IMDB (Movie Reviews)")
amazon_results = evaluate_domain(amazon_test, "Amazon (Product Reviews)")
tweets_results = evaluate_domain(tweets_test, "Tweets (TweetEval)")

```

6.11 Visualization: Confusion Matrices and Cross-Domain Comparison

```

import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

labels_names = ["Negative", "Neutral", "Positive"]
results_list = [imdb_results, amazon_results, tweets_results]

# Confusion matrix heatmaps
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
for ax, result in zip(axes, results_list):
    sns.heatmap(result["confusion_matrix"], annot=True, fmt="d", cmap="Blues",
                xticklabels=labels_names, yticklabels=labels_names, ax=ax, cbar=False)
    ax.set_title(f"{result['domain']}\nAccuracy: {result['accuracy']:.2%}")
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")
plt.tight_layout()
plt.savefig("/content/drive/MyDrive/fincard_bert_project/confusion_matrices.png", dpi=150)
plt.show()

# Cross-domain comparison bar chart
domains = [r["domain"].split(" ")[0] for r in results_list]
accuracies = [r["accuracy"] for r in results_list]
f1_scores = [r["f1"] for r in results_list]

x = np.arange(len(domains))
width = 0.35
fig, ax = plt.subplots(figsize=(9, 6))
ax.bar(x - width/2, accuracies, width, label="Accuracy", color="steelblue")
ax.bar(x + width/2, f1_scores, width, label="Weighted F1", color="orange")
ax.set_xlabel("Domain")
ax.set_ylabel("Score")
ax.set_title("Cross-Domain Performance Comparison")
ax.set_xticks(x)
ax.set_xticklabels(domains)
ax.set_ylim(0, 1.0)
ax.legend()
plt.tight_layout()
plt.savefig("/content/drive/MyDrive/fincard_bert_project/cross_domain_comparison.png", dpi=150)

```

```
plt.show()
```

6.12 Classical Machine Learning Models (TF-IDF + Naïve Bayes / Logistic Regression / SVM)

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix

tfidf = TfidfVectorizer(max_features=10000, ngram_range=(1,2))
X_train = tfidf.fit_transform(combined_train["text"])
X_test = tfidf.transform(combined_test["text"])
y_train, y_test = combined_train["label"], combined_test["label"]

models = {
    "Naive Bayes": MultinomialNB(),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "SVM (Linear)": CalibratedClassifierCV(LinearSVC(max_iter=2000))
}

results = {}
for name, clf in models.items():
    clf.fit(X_train, y_train)
    preds = clf.predict(X_test)
    acc = accuracy_score(y_test, preds)
    precision, recall, f1, _ = precision_recall_fscore_support(y_test, preds,
average="weighted", zero_division=0)
    cm = confusion_matrix(y_test, preds, labels=[0,1,2])
    results[name] = {"model": clf, "accuracy": acc, "precision": precision,
                    "recall": recall, "f1": f1, "cm": cm}
    print(f"{name}: Accuracy={acc:.4f}, Precision={precision:.4f}, Recall={recall:.4f},
F1={f1:.4f}")
```

6.13 ROC Curve Analysis

```
from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt

y_test_bin = label_binarize(y_test, classes=[0, 1, 2])
class_names = ["Negative", "Neutral", "Positive"]

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
for ax, (name, res) in zip(axes, results.items()):
    y_score = res["model"].predict_proba(X_test)
    for i in range(3):
        fpr, tpr, _ = roc_curve(y_test_bin[:, i], y_score[:, i])
        roc_auc = auc(fpr, tpr)
        ax.plot(fpr, tpr, label=f"{class_names[i]} (AUC={roc_auc:.2f})")
```

```

    ax.plot([0, 1], [0, 1], 'k--', linewidth=0.8)
    ax.set_title(f"{name} – ROC Curve")
    ax.legend(fontsize=8)
plt.tight_layout()
plt.show()

```

6.14 Gradio Live Prediction UI

```

import gradio as gr
import torch

def predict_sentiment(text):
    if not text.strip():
        return "Please enter some text."

    model.eval()
    encoding = tokenizer(
        text, truncation=True, padding="max_length",
        max_length=MAX_LEN, return_tensors="pt"
    ).to(device)

    with torch.no_grad():
        outputs = model(**encoding)
        probs = torch.softmax(outputs.logits, dim=1)[0]
        pred_class = torch.argmax(probs).item()

    labels_map = {0: "Negative", 1: "Neutral", 2: "Positive"}
    confidence = probs[pred_class].item()

    result = f"Prediction: {labels_map[pred_class]}\n\nConfidence: {confidence:.2%}\n\n"
    result += f"(Negative: {probs[0]:.2%} | Neutral: {probs[1]:.2%} | Positive: {probs[2]:.2%})"
    return result

demo = gr.Interface(
    fn=predict_sentiment,
    inputs=gr.Textbox(lines=3, placeholder="Type text here...", label="Enter Text"),
    outputs=gr.Markdown(label="Sentiment Prediction"),
    title="FinCard AI Sentiment Engine – Multi-Domain Sentiment Analysis (BERT)",
    description="A BERT-based sentiment classifier trained across movie reviews, product
reviews, and tweets.",
    examples=[
        ["This movie was absolutely fantastic, best I've seen all year!"],
        ["The product broke after two days, total waste of money."],
        ["It's okay I guess, nothing special about it."],
        ["can't believe how bad the service was today smh"],
    ]
)

demo.launch(share=True, debug=True)

```